



Design (E) 314

Technical Report

Embedded Driver Assistance Systems (E-DAS)

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Abstract

The Embedded Driver Assistance System (E-DAS) is developed to track specified vehicle parameters and alert the driver when unsafe conditions are detected, via the use of warning messages, using various peripheral sensing devices. The system is implemented using an STM32-based microcontroller and incorporates components such as an LMT01 Temperature Sensor, HC-SR04 Ultra-sonic Distance Sensor, MPU-6050 Accelerometer, and various other sensor inputs including ADC (Analog-to-Digital) converter inputs.

Contents

List of Abbreviations	6
List of Figures	7
1 Introduction.....	9
2 System Description	9
3 Hardware Design and Implementation	10
3.1 Hardware Block Diagram and Description of Interaction.....	10
3.2 Power Supply.....	11
3.2.1 5V Regulator	11
3.2.2 3.3V Regulator	11
3.3 LEDs for Status Checking.....	12
3.4 Tactile Push-buttons	12
3.5 Temperature Sensor	13
3.6 Light Sensor.....	13
3.7 Ultra-sonic Distance Sensor	14
3.8 Keypad	15
3.9 O-LED Display.....	15
3.10 MPU-6050 Accelerometer	15
3.11 SD Card Reader	16
4 Software Design and Implementation	17
4.1 Software Block Diagram and Description of Interaction	17
4.2 Ultra-Sonic Sensor Measurement and Calibration	18
4.3 Light sensor measurement and calibration	18
4.4 Temperature measurement.....	19
4.5 Accelerometer interface, measurement and calibration.....	20
4.6 UART Communication and System Protocol	20
4.7 Keypad and tactile push-button debouncing.....	21
4.8 OLED Software Driver.....	22
4.9 UI (User Interface) Menu and FSM (Finite-State Machine) Control and Logic.....	22
4.10 Alarm Condition Handling	23

4.11 Real-Time Clock (RTC)	23
4.12 SD card interface and logging	24
5 Measurements and Results	24
5.1 Power Supply Test.....	24
5.2 Keypad and Buttons Test.....	25
5.3 UART Communications Test.....	25
5.4 OLED and Menu Functioning Test	26
5.5 SD Card Test.....	26
5.6 Temperature Sensor Test.....	27
5.7 Ultrasonic Distance Sensor Test.....	28
5.8 Light Sensor Test.....	28
5.9 Accelerometer Test	29
5.10 Entire System Test	29
6 Conclusions.....	30
6.1 Recommendations for Improvement and Notable Deficiencies.....	30
Appendices	32
A.1 Pin Usage	32

List of Abbreviations

Abbreviation	Definition
ADC	Analog-to-Digital Converter
CSV	Comma-Separated Values
DLPF	Digital Low-Pass Filter
DMA	Direct Memory Access
E-DAS	Embedded Driver Assistance System
EXTI	External Interrupt
FAT	File Allocation Table
FATFS	FatFs (FAT Filesystem Middleware Library)
FIR	Finite Impulse Response
FSM	Finite State Machine
GPIO	General Purpose Input/Output
GPS	Global Positioning System
HAL	Hardware Abstraction Layer
HC-SR04	Ultrasonic Distance Sensor Module
I2C	Inter-Integrated Circuit
IC	Input Capture
ISR	Interrupt Service Routine
LED	Light-Emitting Diode
LMT01	Texas Instruments LMT01 Temperature Sensor
LSB	Least Significant Bit
MCU	Microcontroller Unit
MISO	Master In Slave Out
MOSI	Master Out Slave In
MPU-6050	InvenSense MPU-6050 6-Axis IMU (Accelerometer + Gyroscope)
NSS	Negative Slave Select (SPI Chip Select)
OLED	Organic Light-Emitting Diode
PA	Port A (STM32 GPIO Port A)
PB	Port B (STM32 GPIO Port B)
PC	Port C (STM32 GPIO Port C)
PDD	Project Definition Document
PWM	Pulse-Width Modulation
RTC	Real-Time Clock
SCK	Serial Clock (SPI)
SCL	Serial Clock Line (I2C)
SD	Secure Digital (Card)
SDA	Serial Data Line (I2C)

Abbreviation	Definition
SFH203	Osram SFH203 Photodiode
SPI	Serial Peripheral Interface
SSD1306	Solomon Systech SSD1306 OLED Display Controller
SS	System Specification (PDD requirement prefix)
STM32	STMicroelectronics STM32 ARM Cortex-M4 Microcontroller Family
TIM	Timer (STM32 hardware timer peripheral)
TIC	Test Interface Computer
UART	Universal Asynchronous Receiver-Transmitter
USART	Universal Synchronous/Asynchronous Receiver-Transmitter
UI	User Interface

List of Figures

Figure 1	Hardware Block Diagram	1
Figure 2	5V Regulator Circuit	3
Figure 3	3.3V Regulator Circuit	3
Figure 4	Status LED Circuit	4
Figure 5	Tactile Push-Button Circuit	4
Figure 6	LMT01 Comparator and Temperature Sensor Circuit	5
Figure 7	SFH203 ADC Transimpedance Amplifier Circuit	5
Figure 8	HC-SR04 Voltage Divider Circuit	6
Figure 9	Keypad Configuration Diagram	6
Figure 10	MPU-6050 Accelerometer Wiring Diagram	7
Figure 11	SD Card Reader Logic Flow Diagram	7
Figure 12	Software Block Diagram	8
Figure 13	Ultra-sonic Distance Sensor Logic Diagram	8
Figure 14	Light Sensor Logic and Alarm Check Diagram	9
Figure 15	Temperature Sensor Measurement Logic Diagram	10
Figure 16	MPU-6050 Accelerometer Logic Diagram	11
Figure 17	UART General Communication Logic Diagram	12
Figure 18	Keypad and Tactile-Button Debouncing Logic Diagram	13
Figure 19	OLED Software Driver Logic Diagram	13
Figure 20	Finite State Machine and Menu Logic Diagram	14
Figure 21	Alarm Condition Logic Diagram	15
Figure 22	RTC Soft-Tick Logic Diagram	16
Figure 23	SD Card Logging Logic Diagram	16
Figure 24	Measured 3.3V Regulated Voltage Output	17
Figure 25	Measured 5V Regulated Voltage Output	17
Figure 26	Clean Debounced Tactical Button Press Waveform	18
Figure 27	Noise Handling of Tactical Button Press	18

Figure 28	Noise Handling of Keypad Button Press	18
Figure 29	UART TX Line: Student Number Transmission	18
Figure 30	UART Terminal: @Stat& Command Response	19
Figure 31	I2C SCL and SDA Signals during OLED Update	19
Figure 32	Functional OLED Menu Photographs	19
Figure 33	Logged CSV Data on SD Card	20
Figure 34	Created CSV File on SD Card	20
Figure 35	SPI SCK and MOSI Waveforms	20
Figure 36	SPI CS and SCK Timed Signal	20
Figure 37	LMT01 Temperature Sensor Pulse Measurement	20
Figure 38	HC-SR04 Echo Pulse at 5 cm	21
Figure 39	HC-SR04 Echo Pulse at 10 cm	21
Figure 40	Light Sensor Output at 2.7V LED Supply	21
Figure 41	Light Sensor Output at 3.1V LED Supply	21
Figure 42	MPU-6050 Upside-Down Orientation Test	22
Figure 43	MPU-6050 Stationary Orientation Test	22
Figure 44	Demonstration: Warning Messages and LED Functionality (a-e)	23
Figure 45	Demonstration: OLED Menu Pages (a-c)	23
Figure 46	Demonstration: Subheadings, Data Entry and Measurements (a-c)	24

1 Introduction

Road safety is an important concern, and engineering challenge many vehicles on South African roads lack systems that monitor driver behaviour and environmental conditions, increasing the likelihood of unsafe driving and accidents. This project addresses road safety through the design and implementation of a compact, low-cost Embedded Driver Assistance System (E-DAS). The system is built around the STM32F411RE microcontroller and uses an ultrasonic sensor for obstacle detection, an accelerometer for motion and impact monitoring, a light sensor for ambient light measurement, and a temperature sensor for environmental monitoring. Driver feedback is provided through an interactive Organic Light-Emitting Diode (OLED) display and Light-Emitting Diode (LED) indicators. Sensor data is also stored on a Secure Digital (SD) card for later analysis, with additional support for Universal Asynchronous Receiver-Transmitter (UART) communication and Real-Time Clock (RTC) functionality. The aim of the project is to design, implement, and test an embedded system that combines sensing, data processing, user feedback, and data logging into a single platform. This report presents the hardware design, software implementation, experimental results, and overall system performance.

2 System Description

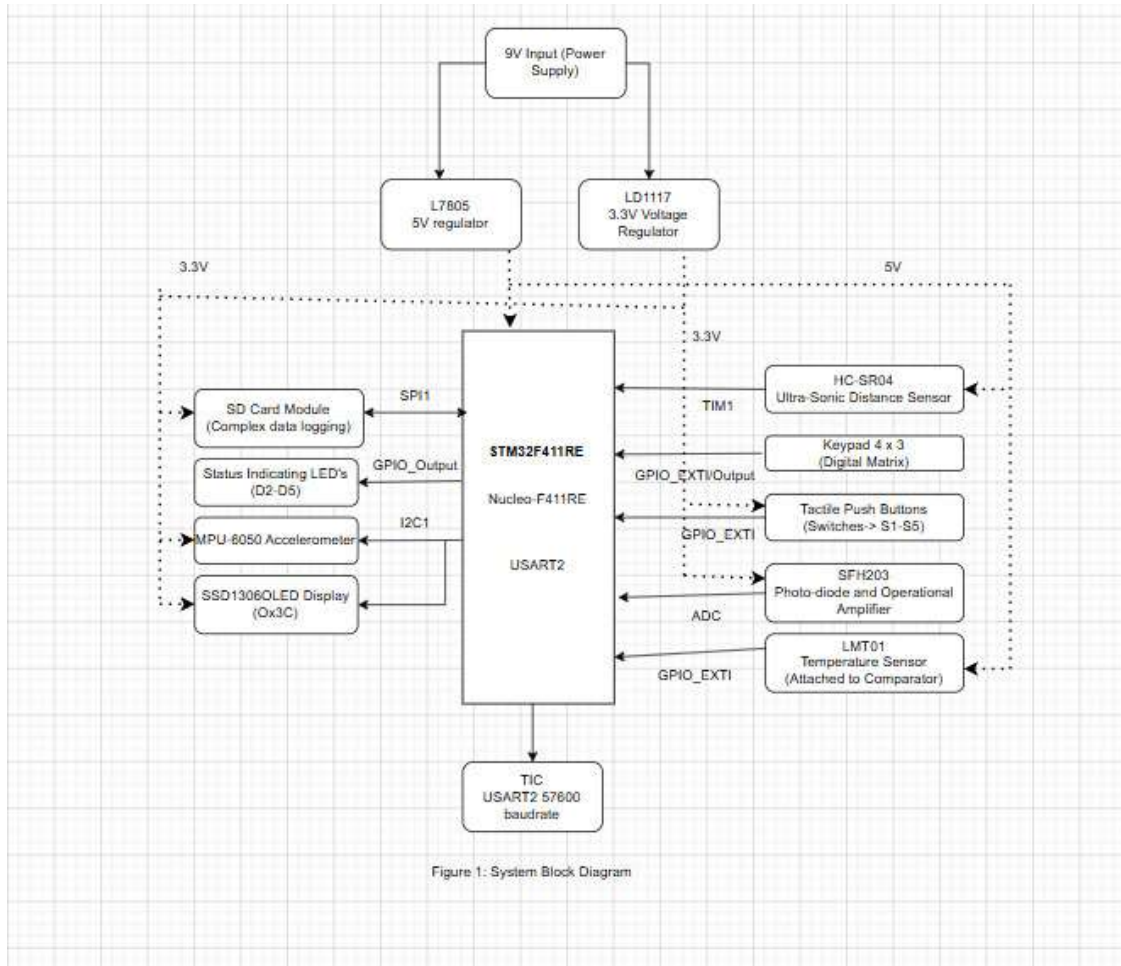
The E-DAS (Embedded Driver Assistance System) is designed to be a self-powered embedded system, with the STM32F411RE acting as the primary processing unit, used to respond to multiple sensors, digital-input devices, output peripherals, communication modules (UART, I2C, SPI) and provide real-time feedback to the driver with minimal latency. The system is powered by a 9V input source which is regulated: to a 3.3V rail supplying the MCU, O-LED display, Indicator LED's and different low-voltage peripherals; and a 5V rail used to supply higher voltage peripherals like the (HCSR04) Ultrasonic Distance Sensor. These regulated voltages ensure that components operate within their optimum voltage range and are not damaged.

The E-DAS system measures the distance to nearby obstacles, monitors motion and acceleration, detects surrounding light conditions, and records ambient temperature, which is presented to the driver in real-time via an OLED Display and dedicated indicator LED's, with important measurements and events (Warnings) stored on an SD card for later review. Drivers can interact with the E-DAS system via tactile push buttons and a matrix-based keypad, which trigger by various GPIO pins on the MCU, allowing navigation of menus and adjustment of operating settings displayed on the OLED, and via UART-telemetry requests. This is useful for specified testing and demonstrations. The E-DAS system gives feedback in the form of indicator LED flashing, Status information and Measurement Information via OLED displays and UART telemetry. UART-based TIC provides a serial communication channel for debugging, command input and system monitoring from an external computer, such as used in demonstrations.

3 Hardware Design and Implementation

3.1 Hardware Block Diagram and Description of Interaction

In the above Figure 1, the hardware sub-systems of the E-DAS are illustrated. A 9V power supply is supplied to the board and regulated via the L7805 to 5V and LD1117 to 3.3V which powers both the STM32F411RE and peripherals. The LMT01 temperature sensor communicates via pulse-counting: its comparative op-amp output drives pin PA15, which is configured as an external interrupt on EXTI_15. The HC-SR04 ultrasonic distance sensor is triggered by a digital output on PA1, while its echo return pulse is captured on PA0 configured as TIM2 channel 1 in input-capture mode. Ambient light is measured by an SFH203 photodiode whose current output is converted to a voltage by an MCP602 transimpedance amplifier before being sampled on PB1 through ADC channel ADC1_IN9. Two peripherals share the I2C1 bus: the SSD1306 128×64 OLED display at address 0x3C, and the MPU-6050 accelerometer at address 0x68 (AD0 low). Data logging to an SD card is handled over SPI1, with chip-select on PC15 and the data lines on PB5 (MOSI), PB4 (MISO), and PB3 (SCK), managed through the FATFS middleware. User interaction is provided by five Tactical Push Buttons (S1–S5 on PC0–PC4 with pull-down resistors configured as GPIO inputs) and a 4×3 matrix keypad whose row drive lines are PB14, PC6, PC5, and PA11, with column lines PC8, PB13, and PA12 configured as General Purpose Input Output External Interrupt lines, with internal pull-ups. Finally, four status LEDs (D2–D5) are driven active-low on PA5, PA6, PA7, and PB6 as standard GPIO outputs. All sensor data and status reports are sent to the test interface (TIC) over USART2, configured at 57 600 baud with even parity and 9-bit word length.

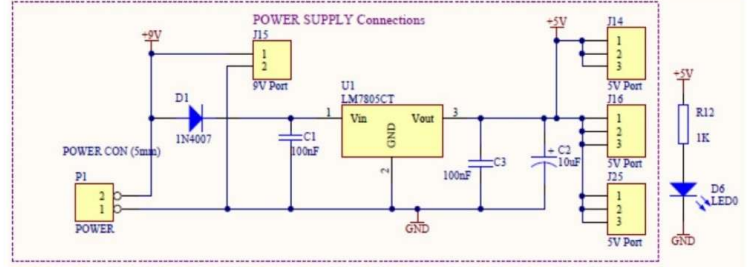


3.2 Power Supply

The E-DAS is supplied with a 9V input voltage. The board regulates this voltage to 5V to power the MCU and 3.3V to power peripherals as seen in as shown in Figure 1.

3.2.1 5V Regulator

Figure 2: 5V Regulator Circuit

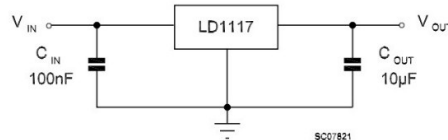


The 5 V rail is powered by an L7805 Linear Voltage Regulator (using the pre-designed circuit from the PDD), shown in Figure 2. The circuit accepts a DC input value between 7 -> 40 V and supplies a stable 5 V to the STM32 board and specified peripherals. A 1N4007 diode at the input provides reverse-polarity protection, C1 helps smooth out supply noise at the input, and C2 (100nF) with C3 (10µF) in parallel filter the output. An indicator LED and series resistor indicate when the regulated output is present, via a red-light output. The 5V regulator dissipates the input-output voltage difference as heat, thermal performance was checked under worst-case conditions. The minimum usable supply voltage is around 7.7–8.0 V once the regulator dropout and diode drop are accounted for, and the practical upper limit is approximately 13.7 V before thermal dissipation becomes a concern at higher current values.

3.2.2 3.3V Regulator

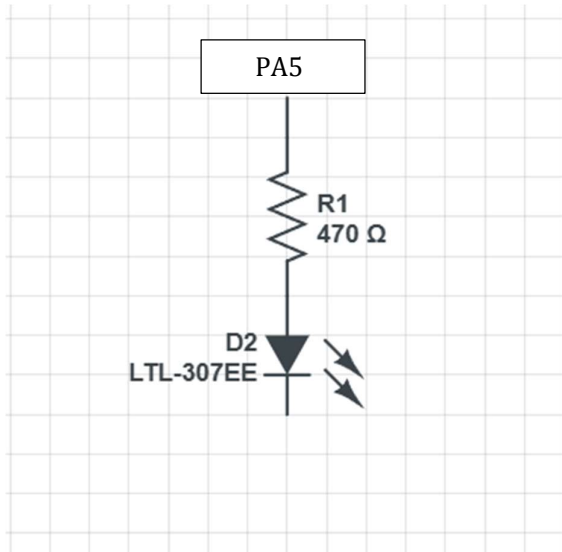
The regulated 5V rail acts as an input supply for the LD1117 3.3V Voltage Regulator, which in turn supplies lower-voltage peripherals and various sub-systems as seen in Figure 1. The design for the regulator comes from the recommended data sheet configuration. The 3.3V regulator is a linear regulator that dissipates excess input voltage as heat; however, a detailed thermal analysis was not considered necessary because the expected dissipation was low. The decoupling capacitors act to reduce the effects of the sudden changes on the 5V input and improve stability and transient response during load changes. Note Figure 3.

Figure 3: 3.3V Regulator Circuit



3.3 LEDs for Status Checking

Figure 4: Status LED



The four LEDs (D2–D5) are powered by the STM32F411RE regulated 3.3V GPIO_Output pins and require a series resistor to limit current, as shown in Figure 4. The recommended LED input current is 20mA, however with a generous use of the 4700hm resistor:

$\frac{V_{in}}{R_{in}} = I_{in}$; we see that we get a value of $3.3/470 = 7\text{mA}$. This value, while under the recommended value of 20mA lowers power consumption, lessens heat dissipation, increases the LED lifespan and reduces stress on drivers and GPIO pins. The brightness in the tests also was sufficient to be seen during demonstrations, which is ideal.

3.4 Tactile Push-buttons

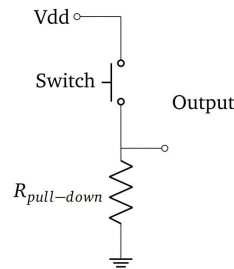


Figure 5: Tactile Push-Buttons

The five tactile buttons are connected to GPIO pins PC0–PC4 and configured as digital inputs with internal pull-down resistors. When a button is pressed it pulls the pin high, giving an active-high logic level, ran on an external interrupt with rising edge trigger. The internal pull-down resistor prevents floating inputs and improves noise immunity, by creating more stability during bounces. Mechanical bouncing was also addressed in code via software rather than an explicit hardware implementation.

3.5 Temperature Sensor

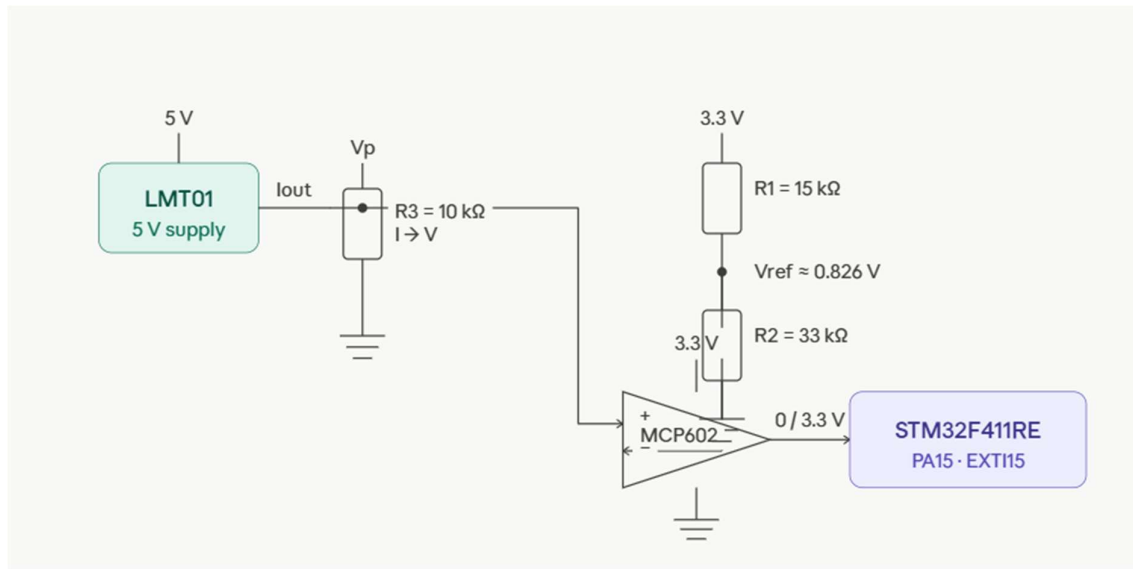


Figure 6: Comparator and Temperature Sensor

Temperature measurement is done by the LMT01 Temperature Sensor, which encodes temperature as a counted series of digital pulses, where the pulse count over a fixed window directly corresponding to the measured temperature. The LMT01 behaves as a current source, so a 10 kΩ resistor (R3) converts its output current to a voltage at Vp. (Based on the datasheet current levels, Vp swings between 0.40 V in the low state and 1.125 V in the high state) which is within the detectable range of the GPIO pins. A MCP602 op-amp in open-loop configuration acts as a comparator, with its supply rails at 3.3 V and 0 V, giving a logic-compatible output. A resistor divider (R1 = 15 kΩ, R2 = 33 kΩ) circuit sets the reference voltage at the inverting input to approximately 0.826 V, which falls cleanly between the two Vp levels, ensuring reliable switching in both directions.

3.6 Light Sensor

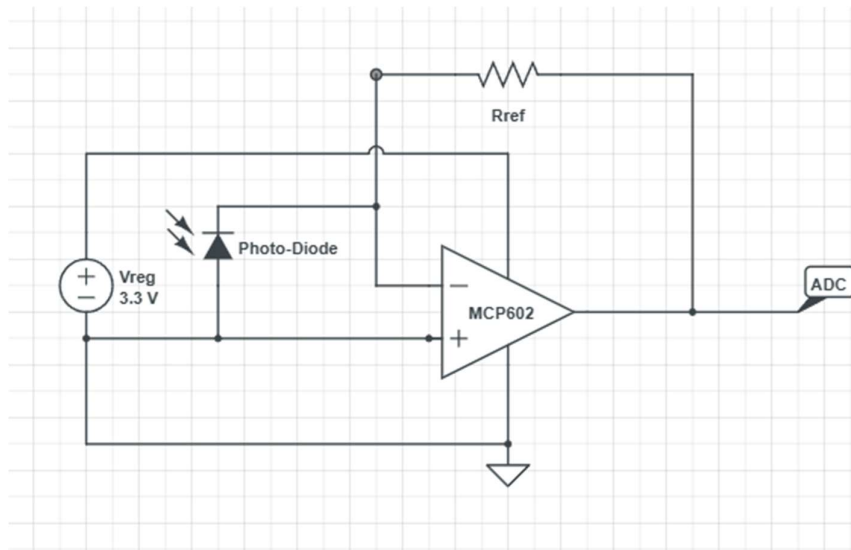


Figure 7: ADC Comparator Circuit

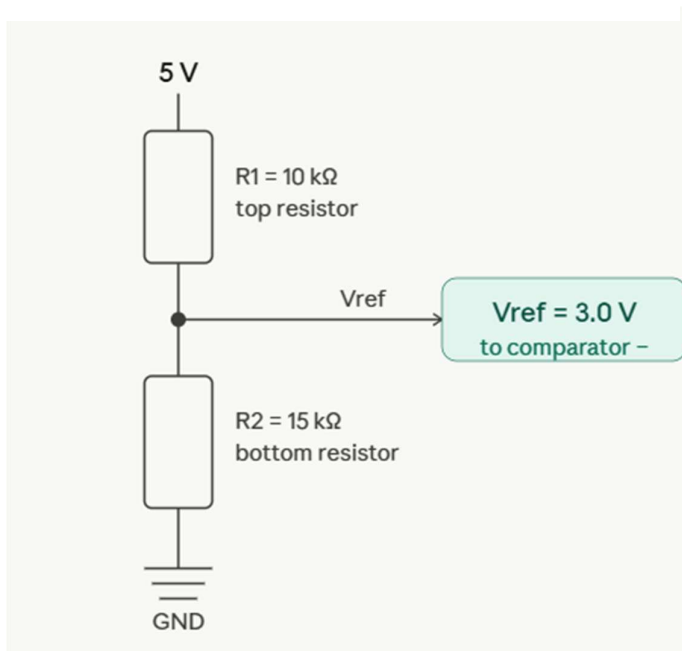
The SFH203 Photodiode generates a current proportional to the intensity of the light hitting it, but since the current is too small to feed directly to the ADC the MCP602 acts as a transimpedance amplifier with the photodiode connected between the supply and the inverting input. The gain is determined by R_f , so it was maximized by the use of a 1Mohm resistor, to ensure that even lower light intensities are observed. The non-inverting input is tied to ground, which forces a virtual ground at the inverting input and ensures the photocurrent flows through R_f . The output voltage is then simply $V_{out} = -I_{ph} \times R_f$, ranging between 0 and 3.3 V, and feeds straight into the ADC.

3.7 Ultra-sonic Distance Sensor

The Ultra-sonic Distance Sensor (HC-SR04) measures distance via emitting an ultrasonic pulse when a trigger signal is applied and measures the time required for the reflected signal to return. The Ultrasonic Distance sensor requires a 5V supply voltage (V_{cc}), however the MPU operates at a maximum of 3.3V and thus we use a voltage divider circuit (Figure 8) to step the Echo signal down to a safe measurable value, using a 15kOhm and 10kOhm resistor. While the data sheets recommend an operating voltage of 5V, the Trig signal is fully functional at 3.3V. The MCU sends a trigger signal by pulling (PA1) trig signal high for 10 μ s, which tells the sensor to fire. Once the burst is sent, the sensor pulls its Echo pin high (PA0), which is captured by TIM2 channel 1 in input-capture mode. From there the distance calculated follows as:

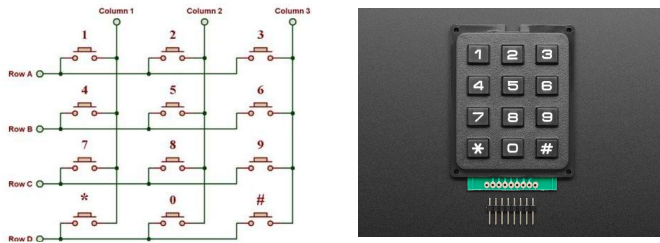
$$\text{Distance} = (\text{Difference} \times 0.034) / 2 /$$

Figure 8: Voltage Divider Circuit



3.8 Keypad

Figure 9: Keypad Configuration



A 4×3 matrix keypad was implemented using row-and-column scanning, this required seven GPIO connections in total. Four GPIO pins were assigned as row outputs (PB14 (Row 1), PC6 (Row 2), PC5 (Row 3), and PA11 (Row 4)), while three pins were configured as column inputs (PC8 (Column 1), PB13 (Column 2), and PA12 (Column 3)) with rising-edge EXTI capability and internal pull-down resistors enabled. Additional series resistors were deemed unnecessary because the MPU could operate safely at these current levels. In idle operation, all four row lines are held at a logic HIGH level. When a button is pressed, the selected row becomes physically connected to a column line, forcing that column input to transition from LOW $\rightarrow$ HIGH and instantly triggering the associated EXTI interrupt.

3.9 O-LED Display

The O-LED panel is a 0.91-inch graphical panel with a pixel resolution of 128×32 , and is used to visually provide feedback, warning messages and direct menu interactions (menu states, alarm notifications, diagnostic feedback). The module itself is controlled by an SSD1306 driver IC, and exchanges data via I2C channels with the STM32F411RE where PA8 is the clock line (I2C_SCK1) and PA9 carries serial data (I2C_SCL1) with a bus speed of 100kHz. The display receives a regulated 3.3V from the board itself, which is well within the recommended 2.5-5.5V operating voltage, as stated in the datasheets. Aleksander Alekseev's stm32-ssd1306 graphics library is used in this project. This package provides routines for screen initialization, text placement, framebuffer manipulation, character rendering, and display refreshing. A 7×10 -pixel font was selected so that the display can accommodate approximately eighteen characters across each line while supporting three readable text rows,

3.10 MPU-6050 Accelerometer

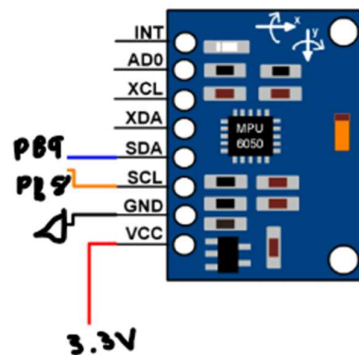


Figure 10: Accelerometer Diagram

The MPU-6050 accelerometer was connected to the STM32F411RE using the same I2C communication bus as the OLED display, which allows both peripherals to share the same serial clock and serial data lines. The peripheral device operates as an I2C slave and provides digital acceleration and angular velocity measurements along three directly orthogonal axes (listed as X, Y, Z in code). The accelerometer module was supplied directly from the regulated 3.3 V on the board to maintain logic-level compatibility with the STM32 microcontroller. Sharing the existing I2C infrastructure also minimised GPIO pin usage, while still providing reliable communication between the MCU, OLED display, and MPU-6050 sensor.

3.11 SD Card Reader

The SD card module communicates with the STM32F411RE over SPI1, with the bus signals mapped to PB3 (SCK), PB4 (MISO), and PB5 (MOSI). PC15 serves as the dedicated chip select pin, set up as a push-pull output running at a high speed. SPI1 runs in master mode with 8-bit frames, MSB-first transmission. CPOL/CPHA = 0/0 with Mode 0 is chosen as it is the default and most widely supported setting. This matches the SPI mode required by the SD card specification. The baud rate prescaler is set to 8, which brings the SPI clock down to 2 MHz from the 16 MHz HSI peripheral clock. This conservative clock rate is because, SD cards must be initialised at less than 400 kHz, and while 2 MHz exceeds that limit for the identification phase, the 500 ms power-up delay added before SD_Init(), allows the card to reach its idle state first. NSS is handled entirely in software, with the FatFs user_diskio_spi layer toggling PC15 around each SPI interaction.

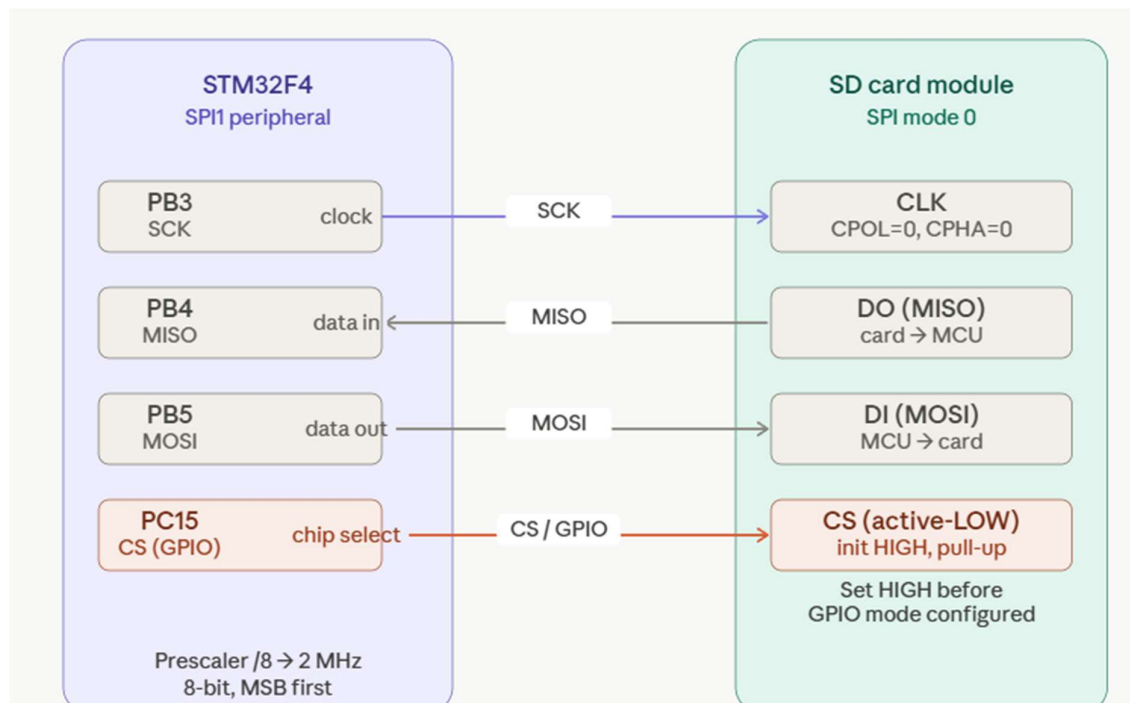


Figure 11: SD Card Reader Logic flow diagram

4 Software Design and Implementation

4.1 Software Block Diagram and Description of Interaction

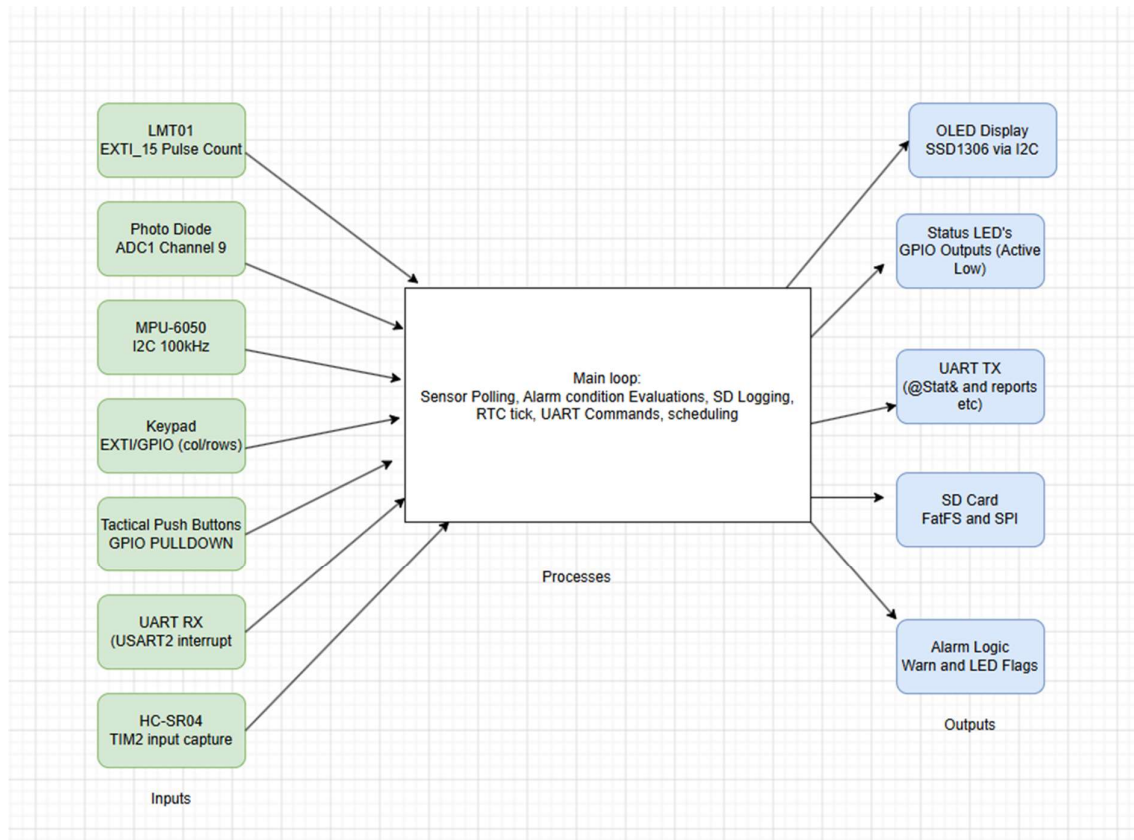


Figure 12: Software Block Diagram

Sensory data from peripherals and user input is directed into a singular while(1) loop, which processes tasks using HAL_Get_Tick() comparisons as a generalized time frame, however each individual process has its own interval constant to ensure that the tasks do not block each other when using the CPU, the outputs of these tasks are driven from the same loop every iteration from start until the program is finished. While the main while loop is running, user defined functions process specific outputs needed by the system, techniques such as the FSM of the OLED display are employed to test various condition-based events.

4.2 Ultra-Sonic Sensor Measurement and Calibration

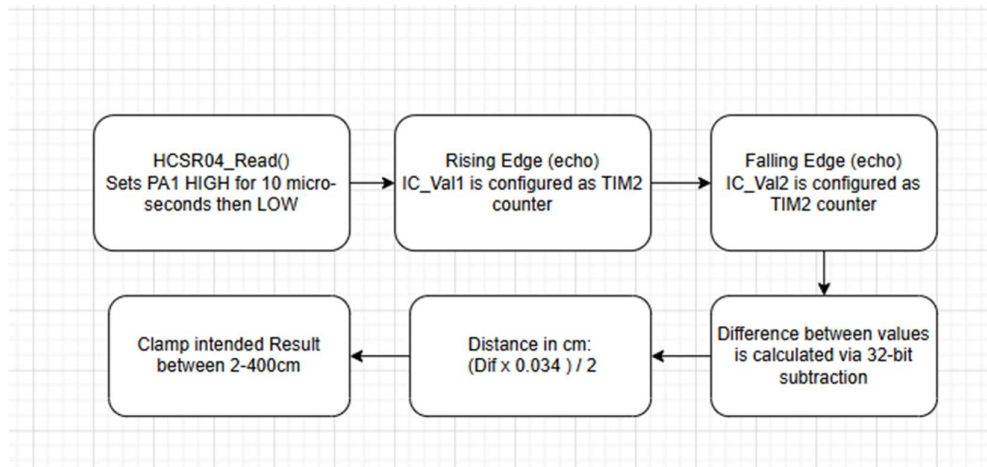


Figure 13: Ultra-sonic Distance Sensor Logic Diagram

HCSR04_Read() sets PA1 HIGH for 10 μ s to send the trigger pulse, then releases it after the dedicated time. TIM2 is configured with a prescaler of 15 (giving 1 μ s resolution at 16 MHz) and operates in input-capture mode on channel 1 (PA0). The first rising edge is captured and stores the value in IC_Val_1 and flips the polarity to falling edge, the second capture (echo pulse) store IC_Val2. The pulse width difference is calculated using 32-bit subtraction to handle counter overflow. The distance is then calculated via $(\text{Difference} \times 0.034) / 2$, which converts the time value into a centimeter distance value for the speed of sound divided by 2 for the trip to the object and back. The final result is clamped between 2-400cm to ensure out-of-range readings are left out.

4.3 Light sensor measurement and calibration

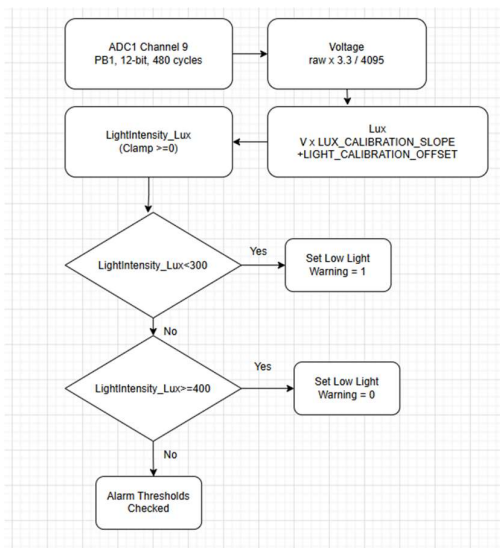


Figure 14: Light Sensor Logic and Alarm Checks

Measure_Light_Intensity() triggers a ADC conversion on channel 9 (PB1) at 12-bit resolution with a 480-cycle sample time, which is used to reduce noise from the high-impedance photodiode circuit. The raw count is scaled to a calculated voltage using $\text{raw} \times 3.3 / 4095$, which is then passed to Convert_Voltage_To_Lux(), which applies a linear calibration: $\text{lux} = \text{voltage} \times \text{LIGHT_CALIBRATION_SLOPE} + \text{LIGHT_CALIBRATION_OFFSET}$, where the slope is 150 lux/V and the offset is zero. The result is clamped to a floor of 0 and stored in LightIntensity_Lux, which the alarm logic compares against the 300 lux and 400 lux threshold value.

4.4 Temperature measurement

The LMT01 outputs temperature as a series of pulses. Start_Temperature_Measurement() configures PA15 as a push-pull output, drives it LOW for 2 ms then HIGH for 3 ms to initiate the conversion, then immediately reconfigures it as a rising-edge EXTI input. Each pulse increments pulse_count in the ISR. After 54 ms (LMT01_DATA_TIME_MS) the main loop reads the count and applies the datasheet formula: $\text{temp} = (\text{pulse_count} / 16.0) - 50$. The raw value is passed through Filter_Temperature(), which is a circular buffer of three readings whose running mean is returned, reducing noise from occasional miscounted or irregular pulses. Readings outside the range 100–4000 pulses are discarded as unrealistic.

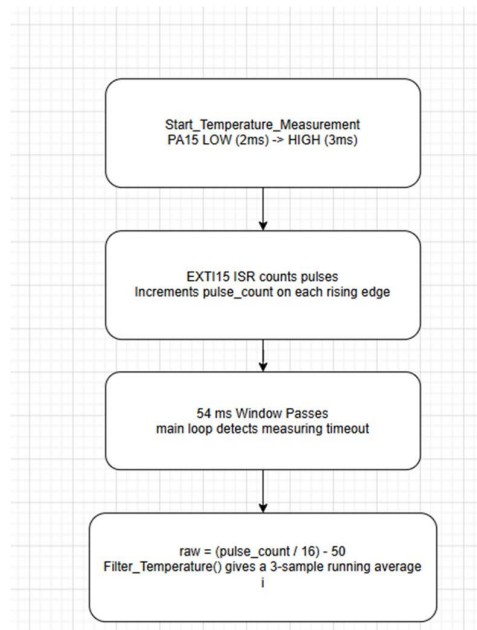
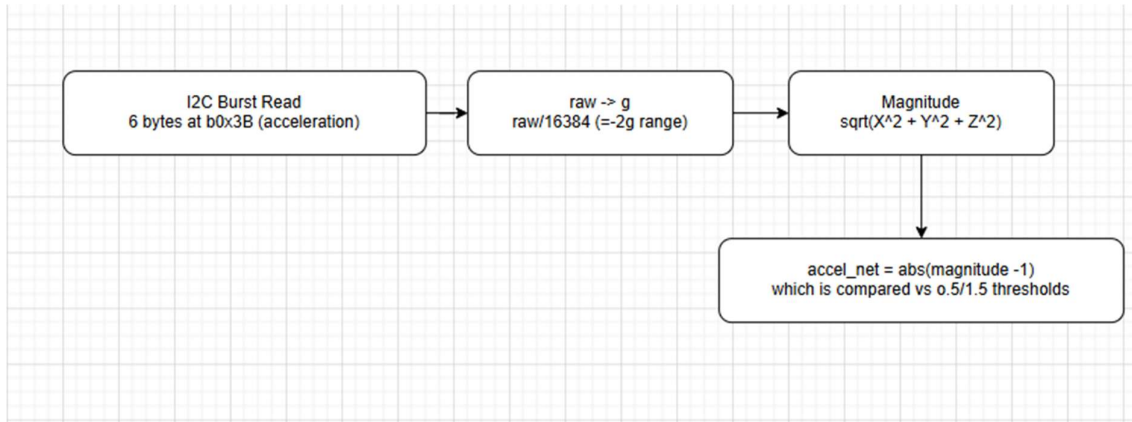


Figure 15: Temperature Sensor Measurement Logic

4.5 Accelerometer interface, measurement and calibration

Figure 16: MPU-6050 Accelerometer Logic



MPU6050_Init() initializes the MPU6050 by writing 0x00 to PWR_MGMT_1, this sets the accelerometer to $\pm 2g$ (16384 LSB/g sensitivity) and the gyroscope to ± 250 dps, enabling a 42 Hz digital low-pass filter. MPU6050_Read_All() performs a 6-byte burst read from register 0x3B every 100ms, assembles the signed 16-bit words and divides by ACCEL_SENSITIVITY to get values in g. MPU6050_Calculate_Net_Accel() then subtracts the 1g gravity component by computing $|\text{magnitude} - 1.0|$, isolating only dynamic acceleration. This accel_net value is then compared against the Unsafe Driving (0.5g) and Impact Detected (1.5g) thresholds in Update_Alarm_Conditions().

4.6 UART Communication and System Protocol

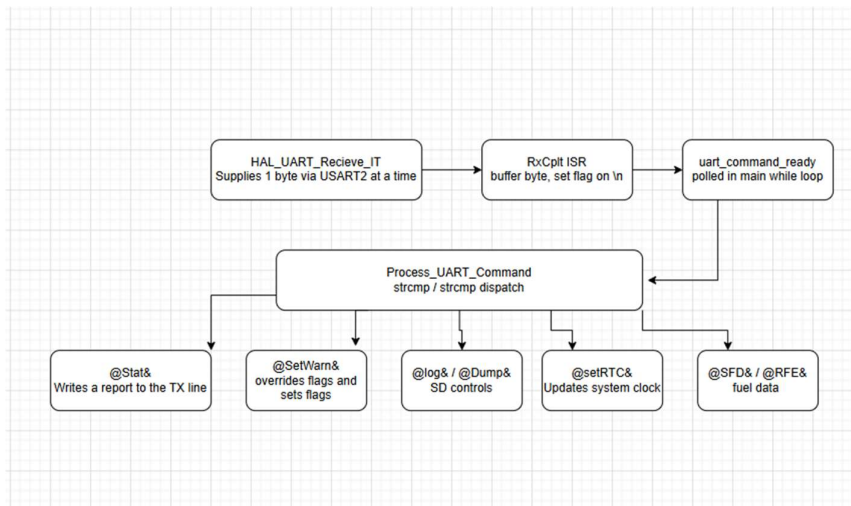


Figure 17: UART General Communication Logic

USART2 is configured at 57600 baud with 9-bit words and even parity. Bytes arrive via the interrupt-driven HAL_UART_Receive_IT, accumulating in uart_rx_buffer sequentially via HAL_UART_RxCpltCallback. When a newline character is detected the uart_command_ready flag is set, and the main loop calls Process_UART_Command() on the next iteration. Commands follow the generalised @...&\n frame format and are dispatched using strcmp/strncmp. Responses are wrapped in the same @...&\n envelope. The @SetWarn command supports per-channel numeric arguments parsed with sscanf, enabling individual warning overrides without requiring a direct menu interaction.

4.7 Keypad and tactile push-button debouncing

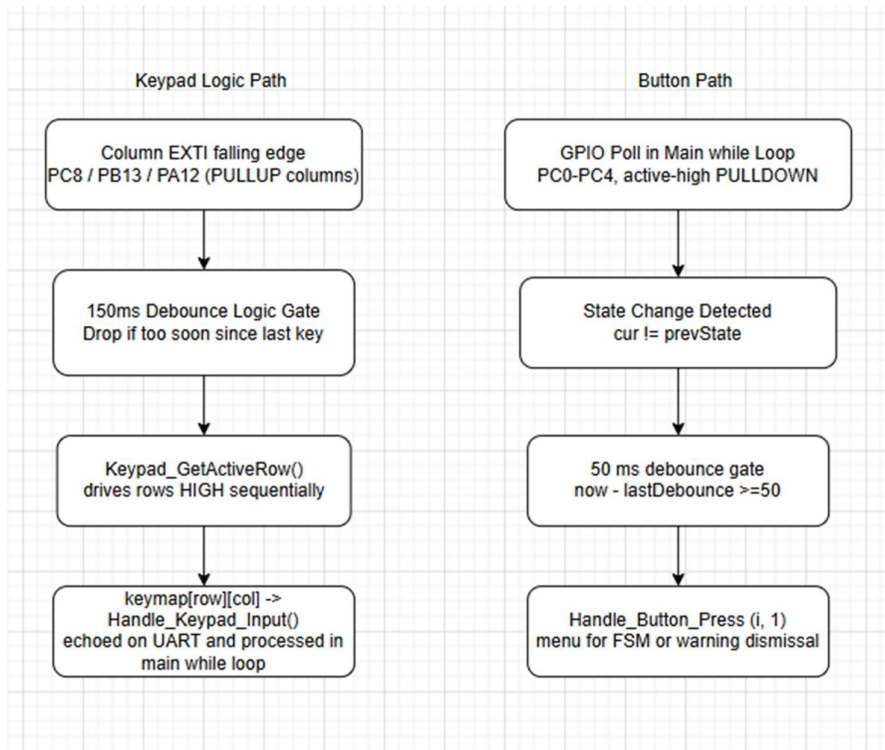


Figure 18: Keypad and Tactile-Button debouncing and button logic

The keypad uses a falling edge trigger on an external interrupt for the three column pins (held HIGH by internal pull-ups). When a key press pulls a column LOW, the EXTI fires and `Keypad_GetActiveRow()` drives each row HIGH in turn, reading back the columns to identify the pressed row. A 150 ms software debounce gate (`KEYPAD_DEBOUNCE_MS`) ensures any interrupts that trigger too early are ignored. The identified `keymap[row][col]` character is stored in `keypad_last_key` and the `keypad_key_ready` flag is set for the main loop. The five tactile buttons are polled directly each loop iteration. A state-change check (`cur != prevState`) combined with a 50 ms timestamp guard (`DEBOUNCE_MS`) prevents mechanical bounce. Only rising edges (`cur == GPIO_PIN_SET`) are forwarded to `Handle_Button_Press()`, ensuring each physical press generates only exactly one event.

4.8 OLED Software Driver

Update_OLED_Menu() is set in 200 ms rate intervals to prevent I2C bus saturation, as the SSD1306 shares the I2C1 line with the MPU-6050 accelerometer. Each update begins with `ssd1306_Fill(Black)` to wipe the internal frame buffer. Specified values are then written using `snprintf` to format sensor values, and `ssd1306_WriteString` is used to render them at fixed pixel positions using the three predefined row offsets (ROW1_Y=0, ROW2_Y=11, ROW3_Y=22). A single call to `ssd1306_UpdateScreen()` at the end flushes the complete 128×32 pixel frame buffer to the display over I2C in one burst, ensuring the screen looks seamless.

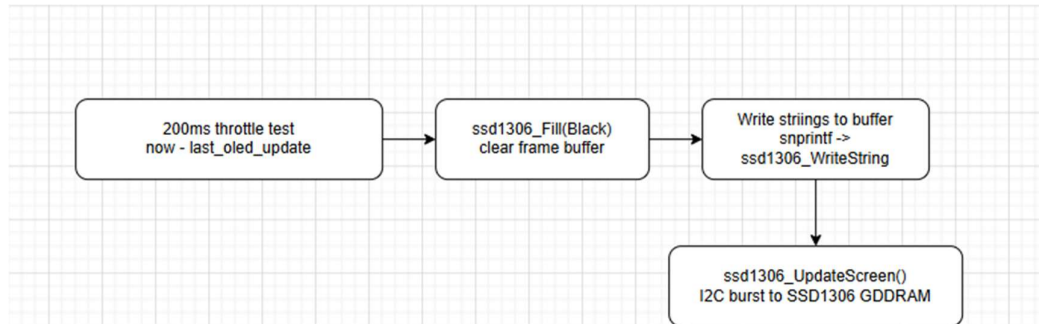
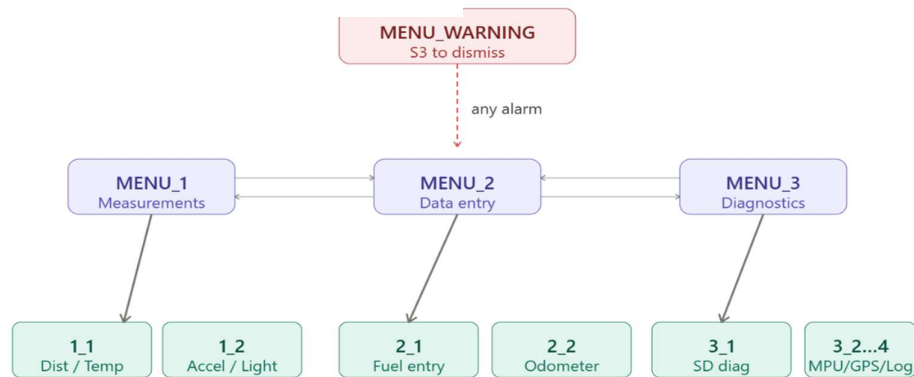


Figure 19: OLED Software Driver Logic

4.9 UI (User Interface) Menu and FSM (Finite-State Machine) Control and Logic

Figure 20: Fine State Machine and Menu Logic



S1/S5: cycle top-level
 S4: enter sub-menu / cycle forward
 S2: back to top-level
 S3: edit/confirm or dismiss warning

The UI is implemented as a finite-state machine using the `MenuState_t` enum definition. `Handle_Button_Press()` evaluates the transitions in a specified priority order: warning dismissal (S3 while in `MENU_WARNING`) is checked first, then edit/confirm mode for the data-entry sub-pages, and finally normal menu navigation. S1/S5 cycle through the three top-level states, S4 enters a sub-menu or steps forward through its pages, S2 returns to the top level, and S4/S5 wrap around at the last sub-page. `MENU_WARNING` can be entered from any state by `Show_Warning()`, which saves the current state in `pre_warn_state` so `Dismiss_Warning()` can restore it to its previous state.

4.10 Alarm Condition Handling

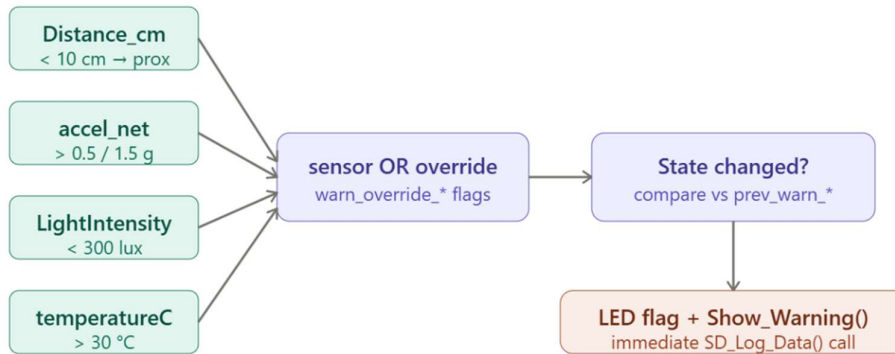


Figure 21: Alarm Logic

Update_Alarm_Conditions() evaluates five independent conditions each loop iteration, where each condition is a logical OR of the real-time sensors from peripherals, exceeding its threshold and the corresponding warn_override_* flag, allowing UART-commanded overrides to simulate any alarm without hardware input. When a condition becomes active its associated LED (D2-D5) is set and Show_Warning() is called to push the menu FSM into MENU_WARNING. After each evaluation, the new state is compared against the cached prev_warn_* value and any transition sets a log_now flag that causes an immediate SD_Log_Data() UART call, satisfying the sub-100 ms logging requirement on warning events.

4.11 Real-Time Clock (RTC)

RTC_SoftTick() is called every main loop iteration and returns immediately if less than 1000 ms have elapsed since the last tick, to ensure measurements are only taken per 1s. When the interval is reached, it increments rtc_sec and carries upward through minutes, hours, and days using a days_in_month[] lookup table. February is extended to 29 days when the year is a leap-year (%4==0 && %100!=0 or %400==0). The current date and time can be overridden at any point via the @SetRTC YYYY/MM/DD;HH:MM:SS& UART telemetry command, which also resets last_rtc_tick to prevent an unpredicted tick.

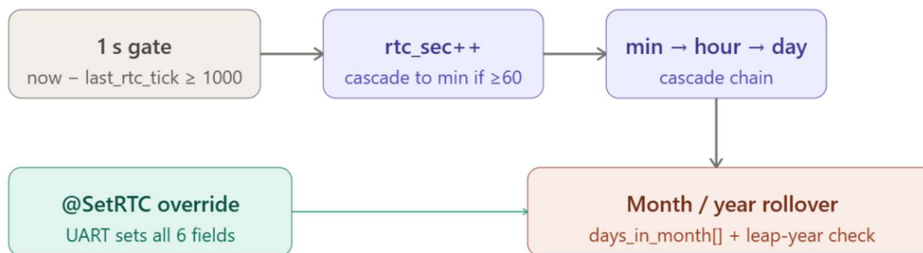


Figure 22: RTC logic

4.12 SD card interface and logging

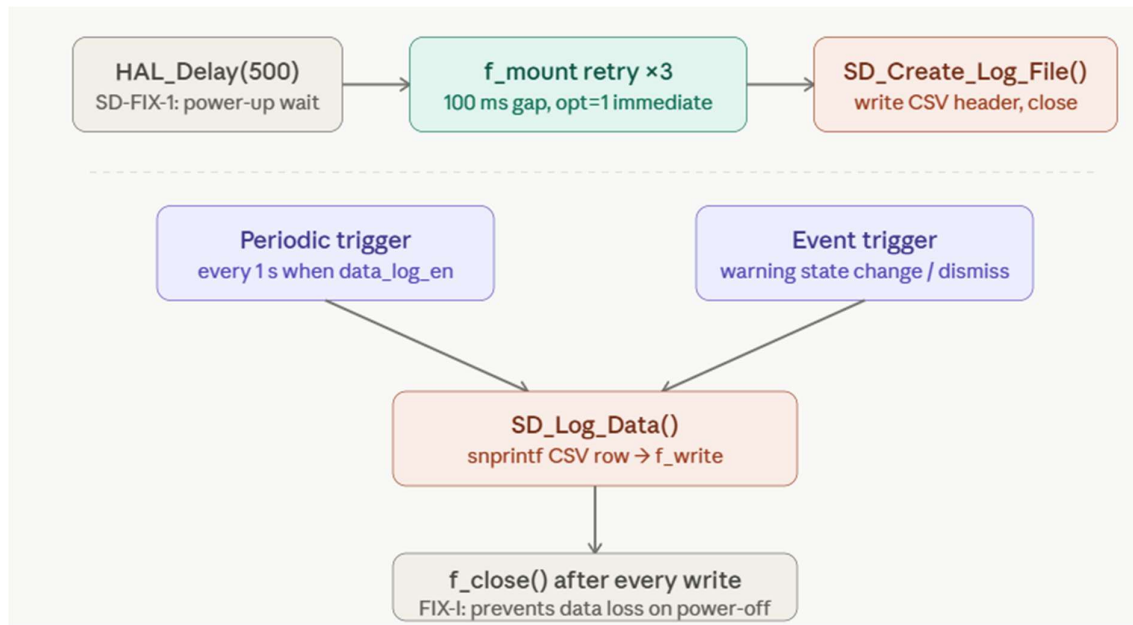


Figure 23: SD Card Logic

At startup, a 500 ms delay gives the SD card time to reach its idle state before `SD_Init()` attempts `f_mount`. The mount is retried up to three times with 100 ms gaps to handle specific cards that need additional clock cycles. On success, `SD_Create_Log_File()` opens `log.csv` with `FA_CREATE_ALWAYS` and writes the CSV header. Logging has two triggers: a periodic 1-second interval controlled by `data_log_en`, and an immediate write whenever any warning state transitions, set or cleared, and is detected by comparing each condition against its `prev_warn_*` cache. In both cases `SD_Log_Data()` formats a single CSV row using `sprintf`, opens the file in append mode, calls `f_write`, and immediately calls `f_close()`.

5 Measurements and Results

5.1 Power Supply Test

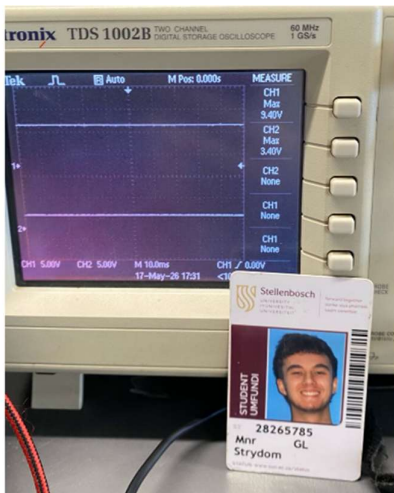


Figure 24: Regulated 3.3V

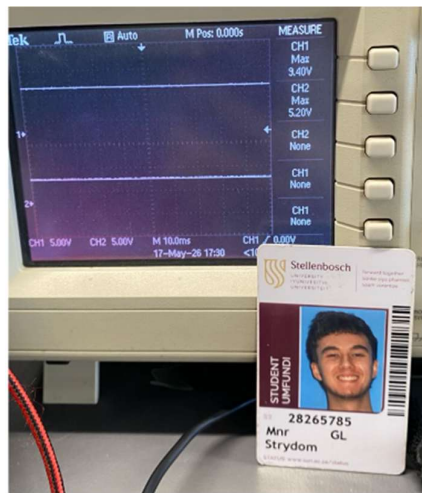


Figure 25: Regulated 5V

As shown by figure 24 and 25 the values for the regulated 3.3V and 5V inputs are within 5% of ideal which satisfied the SS1 requirement as stated in the PDD. Abs (Videal - Vreal / Videal). The 3.3V line, $3.4 - 3.3 / 3.3$, was within 3.03% of the ideal value and the 5V line, $5.2 - 5 / 5$, was within 3% of the ideal value.

5.2 Keypad and Buttons Test

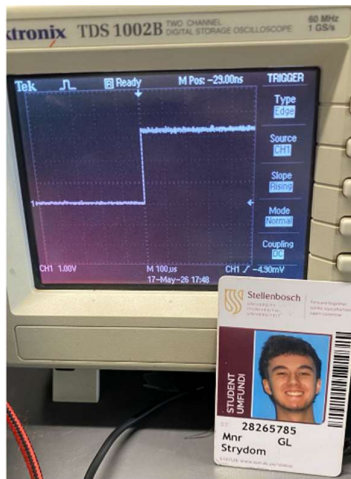


Figure 26: Clean Debounced Tactical Button Press



Figure 27: Noise Handling of Tactical Button Press

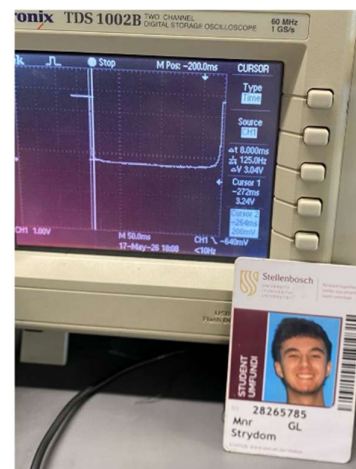


Figure 28: Noise Handling of Keypad Button Press

Oscilloscope measurements showed that the digital software solution for the debouncing was effective with a transition of LOW to HIGH on the Tactical button Press occurring well within the range of the allocated debouncing time window and having a clear logical LOW to HIGH waveform. The keypad also had an effective transition from HIGH to LOW, with the logical HIGH for each of the buttons being represented by the MPU's 3.3V. Additional testing with repeated irregular presses and prolonged holds demonstrated predictable behavior, as every interaction generated exactly one recognized input without duplicate responses. Measurements on the keypad matrix verified proper functionality. During a key activation, the controlled row line transferred a signal to the matching column path, creating a detectable logic change at the controller input. By scanning rows sequentially and monitoring columns through interrupt-based sensing, the software accurately identified the selected key location.

5.3 UART Communications Test



Figure 29: UART sending student number via TX line

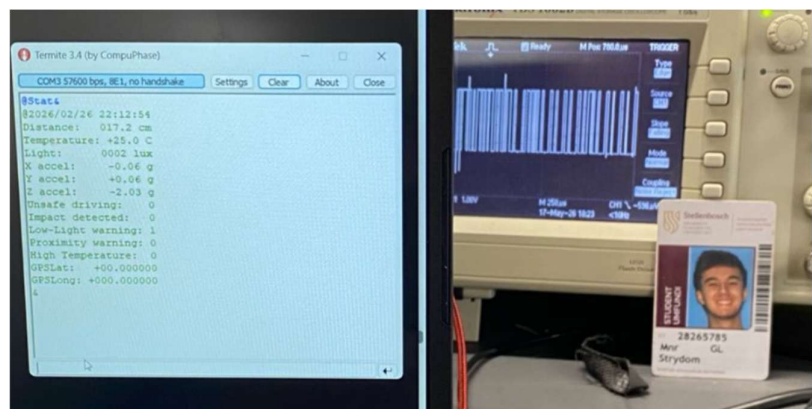


Figure 30: UART sending @Stat& command

The UART communication line was set at a baud rate of 57600 baud with an 8-bit frame structure, even parity checking, and a single stop bit in accordance with the system requirements. The microcontroller correctly sent the identification message *28265785# to the terminal interface of the oscilloscope. A status request command was then issued from the computer, prompting the embedded system to return formatted diagnostic information containing live measurement values and alert conditions. The displayed terminal data confirmed successful encoding and interpretation of ASCII-based messages, which was further confirmed by the oscilloscope reading. The bit period measured approximately 17.26 microseconds which when converted is 57937 baud, the difference could be explained by noise or wire interference.

5.4 OLED and Menu Functioning Test

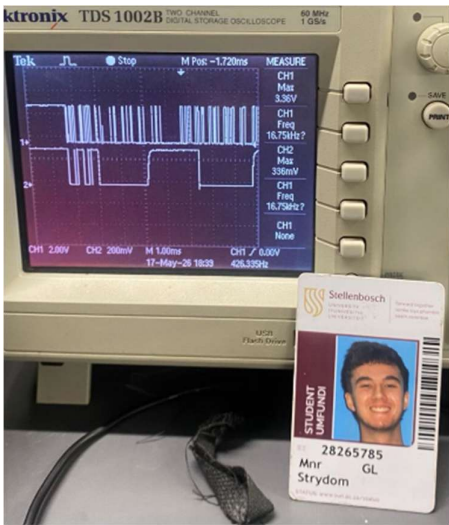


Figure 31: I2C communication via SCL and SDA during OLED Updates



Figure 32: Functional OLED Menu

The OLED was tested during states of alarms, changing menus, on power-up and while operating real-time with interactive data. Oscilloscope probing of the serial clock and data connections confirmed proper digital signaling activity during screen refresh operations, verifying successful data transfer between the embedded controller and the display hardware. Measurement readings and navigation information updated continuously without graphical errors, flickering, or system lockups, demonstrating stable SSD1306 controller functionality and dependable communication with the microcontroller through the I2C bus, this shows functional SDA and SCL lines.

5.5 SD Card Test

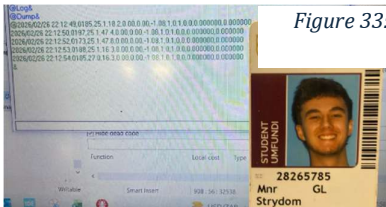


Figure 33: logged CSV data



Figure 34: created CSV file

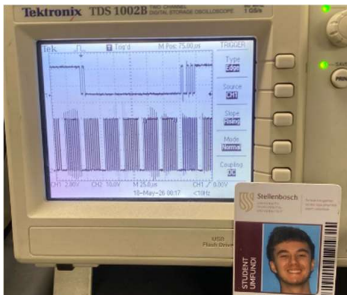


Figure 35: SCK and MOSI signals

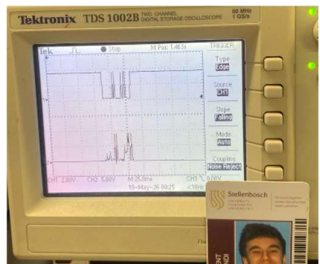
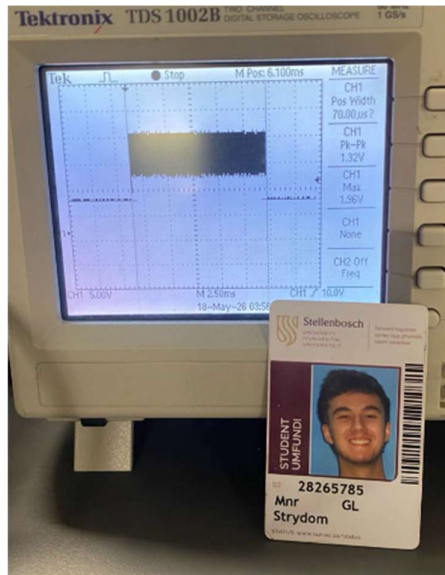


Figure 36: CS and SCK timed signal

The SD card was tested via monitoring serial peripheral communication activity (SPI signals), confirming successful generation of a storage file (CSV), and checking the validity of the recorded information. The oscilloscope showed that the SPI bus is fully operational, including continuous data transfer, clock pulses, and device-enable control during write operations. The oscilloscope also showed that the sampling requirement of 1 second was met.

5.6 Temperature Sensor Test



Pulse Measurement

The LMT01 Temperature Sensor was analyzed by accessing the generated pulse sequence and timing characteristics on an oscilloscope and from that calculating an expected room temperature. The microcontroller counted pulses within a fixed interval and converted the result into a temperature estimate via a linear approximation method. Temperatures higher than 30C were tested to trigger alarm conditions. The comparator attached to the LMT01 ensured that the pulses remained stable and within a - 0.16V and 3.36V range, as demonstrated by the 1.96V in Figure 37. The measured cycle duration was 12.23 microseconds which is relatable to the 11.56 micro-second expected value from a 88 kHz signal of the LMT01 (as stated in the datasheets).

5.7 Ultrasonic Distance Sensor Test



Figure 38: Voltage measurement at 5cm for echo pulse

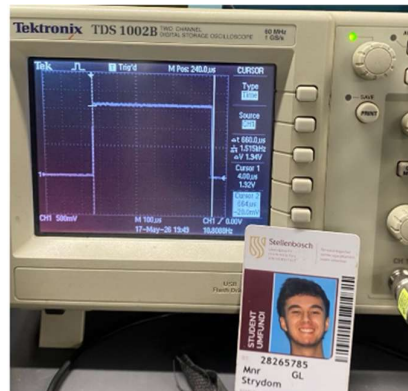


Figure 39: Voltage Measurement at 10cm for echo pulse

From the oscilloscope measurements at figure 38 and 39 we see time intervals of 320 microseconds and 660 microseconds accordingly. Using the implemented relation of $D_{cm} = \Delta t \times 0.034 / 2$, (divided by 2 for the return sound wave) where the change in time is measured as microseconds, we see that the estimated distance 5.44cm and 11.22cm which is within 2 cm of the actual value, despite outside circumstances like movement or noise. The HC-SR04 measures distance by transmitting a short burst of 40 kHz sound waves and timing how long the reflected signal takes to return. The microcontroller initiates the measurement using a 10 μ s trigger pulse, after which the sensor drives its Echo output high until the reflected sound is detected. A hardware timer operating in input-capture mode records the rising and falling edge timestamps of the Echo signal, allowing the pulse duration to be determined with microsecond precision.

5.8 Light Sensor Test

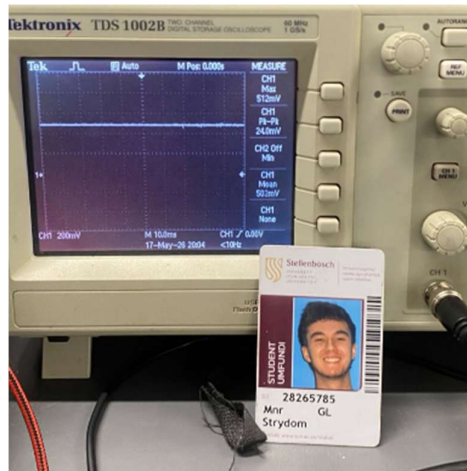


Figure 40: Light Sensor Output for 2.7V LED

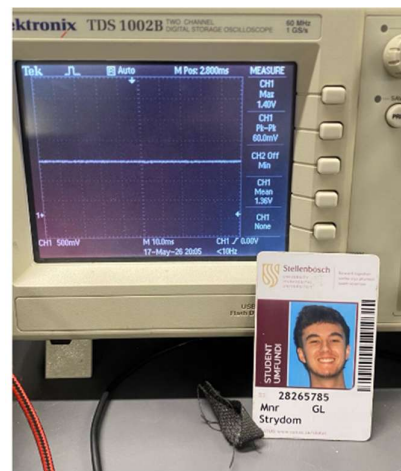


Figure 41: Light Sensor Output for 3.1V LED

According to the datasheets: $ADC = V_{ADC} \times 4095 / 3.3$, hence my values for ADC would be 636 and 1738, which according to the calibration curve would clear the low light warning. When the LED supply increased to 3.1 V, the sensor output rose to approximately 1.4 V, producing a calculated brightness value of about 189 lux. Using these measured operating points, a first-order linear calibration relationship was derived to convert the analogue sensor voltage into lux values. The results confirmed that the circuit produced a higher output voltage and correspondingly larger lux value under increased illumination, validating correct photodiode response, ADC conversion, and calibration behavior.

5.9 Accelerometer Test



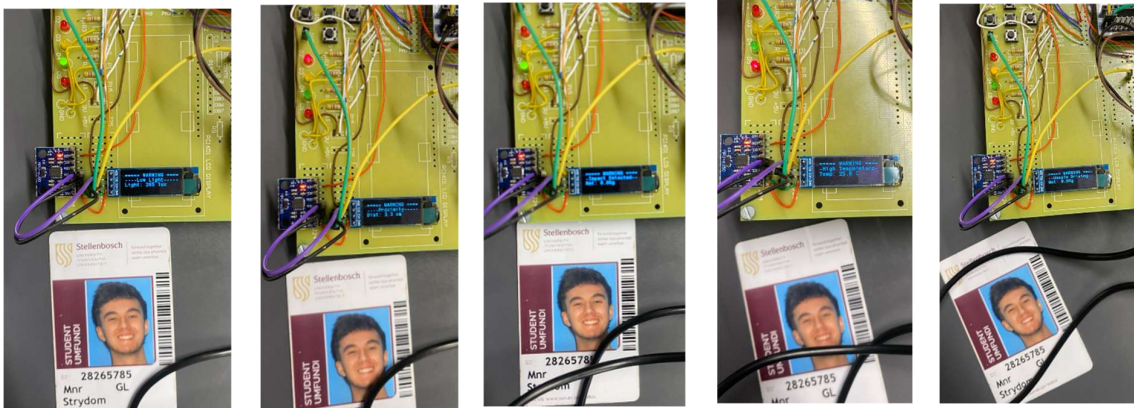
Figure 42: Upside Down Accelerometer test



Figure 43: Stationary Accelerometer test

The accelerometer was tested at various orientations to validate gravity and logic calibration, measured acceleration magnitude was approximately 0.07 g, confirming successful offset correction and effective removal of the static gravitational component from the Z-axis measurement. When the board was rotated upside down by 180°, the measured Z-axis acceleration was approximately -1.93 g, which closely matches the expected theoretical value of -2 g after gravity compensation. These results confirmed accelerometer scaling, reliable orientation sensing, and correct MPU6050 operation. The system also successfully triggered the Unsafe-driving and Impact-warning conditions when acceleration thresholds were exceeded ($>1.5g$ and $>1g$).

5.10 Entire System Test



Figures 44 (a-e): Demonstrating warning messages and LED functionality

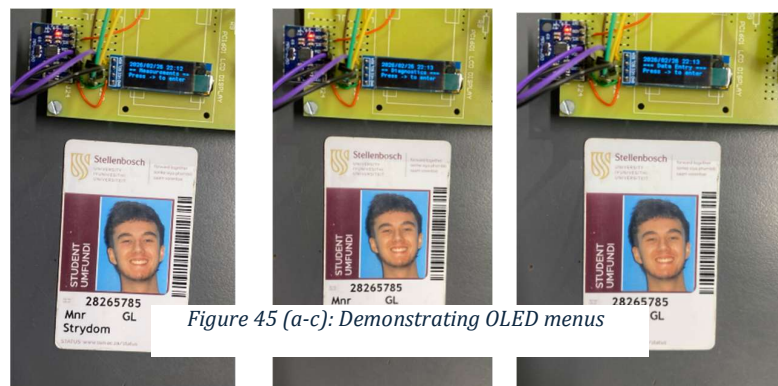


Figure 45 (a-c): Demonstrating OLED menus



Figure 46(a-c): Demonstration of subheadings, data entry capability and continuous measurements

The final E-DAS system is portrayed in figure 46c which includes all allocated peripherals (excluding the GPS system), figures 44 (a-e) demonstrate warning messages and various peripheral functionality on the OLED display, while figures 45 (a-c) demonstrate the various menus available. Figures 46a and 46b show the capabilities of using the keypad for data inputs and for specific user-based commands. The accelerometer only appears after successful handshake and valid responses over the I2C interface, while the storage indicator is enabled once the memory card is properly mounted and SPI communication is verified through the FAT file system layer. All appropriate functionality is displayed in terms of peripherals and previous functionality is demonstrated in the previous sections.

6 Conclusions

6.1 Recommendations for Improvement and Notable Deficiencies

The E-DAS, while functional, had a few notable flaws. Optimizing communication protocols (I2C, SPI, and UART) to reduce bus overhead and unnecessary polling would improve both performance and energy efficiency. Preferring lower power states, like active LOW, could promote using less power and increase the lifespan of components, especially if there is an option in rapid sampling peripherals like the Ultrasonic Distance Sensor or the Temperature sensor. Using less wires in favor of a more well-designed PCB board could reduce waste and improve efficiency and build-time. Implementing data logging compression could allow the system to store data more effectively without causing damage to components. Using peripherals with multiple functionalities would also use less components and be more affordable from a business perspective. Calibration of components could also be done more effectively instead of basic linear extrapolation, which would increase the accuracy of the measurements.

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Appendices

A.1 Pin Usage

Pin	Peripheral Function /	Direction	Notes
Port A — GPIOA			
PA0	TIM2_CH1 — HC-SR04 Echo	Input	<i>Timer input capture, rising then falling edge</i>
PA1	HC-SR04 Trigger	Output	<i>10 μs pulse to fire ultrasonic burst</i>
PA5	LED D2 — Proximity warning	Output	<i>Active-low; GPIO_PIN_RESET = ON</i>
PA6	LED D3 — Impact / unsafe driving	Output	<i>Active-low; solid ON (impact) or 500 ms flash (unsafe)</i>
PA7	LED D4 — Low-light warning	Output	<i>Active-low; 500 ms flash when active</i>
PA11	Keypad Row 4 (bottom)	Output	<i>Driven high to scan row 4; idle LOW</i>
PA12	Keypad Column 3 — EXTI12	Input PULLUP / EXTI falling	<i>Shared EXTI15_10 IRQ; detects key press in col 3</i>
PA15	LMT01 Temperature pulse — EXTI15	Bidirectional (OPP then EXTI)	<i>2 ms LOW reset pulse, then rising-edge EXTI counts pulses; count $\rightarrow$ °C</i>
Port B — GPIOB			
PB1	Photodiode ADC — ADC1_IN9	Analog input	<i>12-bit ADC, 480-cycle sample; voltage $\rightarrow$ lux via linear calibration</i>
PB3	SPI1_SCK — SD card clock	AF Output	<i>SPI1 alternate function</i>
PB4	SPI1_MISO — SD card data in	AF Input	<i>SPI1 alternate function</i>
PB5	SPI1_MOSI — SD card data out	AF Output	<i>SPI1 alternate function</i>
PB6	LED D5 — High-temperature warning	Output PP	<i>Active-low; 500 ms flash when active</i>
PB13	Keypad Column 2 — EXTI13	Input PULLUP / EXTI falling	<i>Shared EXTI15_10 IRQ</i>

Pin	Peripheral Function /	Direction	Notes
PB14	Keypad Row 1 (top)	Output PP	<i>Driven high to scan row 1; idle LOW</i>
Port C — GPIOC			
PC0	Button S1 — UP	Input PULLDOWN	<i>Active-high; 50 ms debounce</i>
PC1	Button S2 — LEFT (Back)	Input PULLDOWN	<i>Active-high; 50 ms debounce</i>
PC2	Button S4 — RIGHT (Next)	Input PULLDOWN	<i>Active-high; 50 ms debounce</i>
PC3	Button S5 — DOWN	Input PULLDOWN	<i>Active-high; 50 ms debounce</i>
PC4	Button S3 — MIDDLE (Select / Dismiss)	Input PULLDOWN	<i>Active-high; dismisses warnings, confirms data entry</i>
PC5	Keypad Row 3	Output PP	<i>Driven high to scan row 3; idle LOW</i>
PC6	Keypad Row 2	Output PP	<i>Driven high to scan row 2; idle LOW</i>
PC8	Keypad Column 1 — EXTI9_5	Input PULLUP / EXTI falling	<i>EXTI9_5 IRQ; detects key press in col 1</i>
PC15	SD card SPI CS (chip select)	Output PP	<i>Active-low; deasserted HIGH at boot; SPI_NSS_SOFT</i>
I²C1 — shared bus (SDA/SCL not user-configurable in CubeMX table)			
I ² C1	MPU-6050 IMU (addr 0x68 / 0xD0)	I ² C master	<i>Accel ±2g, gyro ±250 dps, DLPF 42 Hz, 125 Hz sample rate</i>
I ² C1	SSD1306 OLED 128×32	I ² C master	<i>Shared bus with MPU-6050; 100 kHz</i>
USART2			
USART2	PC UART — command / telemetry	TX + RX	<i>57600 baud, 9-bit, even parity; interrupt-driven RX; commands: @Stat&, @Log&, @Dump&, @CLF&, @SetWarn&, @SFD&, @RFE&, @SetRTC&</i>
Timers			

Pin	Peripheral Function /	Direction	Notes
TIM2	HC-SR04 echo input capture	IC CH1	<i>Prescaler 16-1 $\rightarrow$ 1 μs ticks; measures echo pulse width $\rightarrow$ distance</i>